Unit 57 Contribution of analytical chemistry to our society

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

N1 Refer to page T88.

N2 Refer to page T88.

57.1 Introduction

Analytical chemistry serves the needs of many fields. For example, N1

- food and drugs are analyzed in quality control tests; N2
- air quality is often evaluated by testing for suspected pollutants;
- forensic chemistry can provide legal evidence;
- in medicine, analytical chemistry is the basis for clinical laboratory tests which help doctors diagnose diseases and chart progress in recovery.

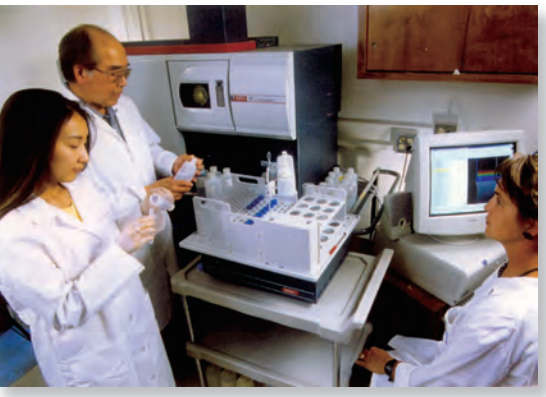


Fig. 57.1 Chemists preparing blood and urine samples for analysis

57.2 Food analysis

Both traditional analytical methods (such as gravimetric and volumetric analyses) and modern instrumental analytical methods play an important role in food analysis.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Based on previous knowledge learnt from this course, suggest a method for each of the following analyses.

1 Determining the salt content in brine

Extension

2 Determining the calcium content in food

3 Determining the phosphorus content in food

4 Determining the acidity of fruit juices

5 Determining the ethanoic acid content in vinegar

6 Determining the sulphur dioxide content in wine

1 Titrating with aqueous solution of silver nitrate / precipitation using an aqueous solution of silver nitrate in gravimetric analysis
2 Precipitation using oxalate ions in gravimetric analysis
3 Treating the sample with molybdenum blue to obtain a blue-coloured species and measuring the absorbance of the species using a colorimeter
4 Titrating the acid with standard aqueous solution of sodium hydroxide
5 Titrating the ethanoic acid with standard aqueous solution of sodium hydroxide
6 Titrating the sulphur dioxide with standard aqueous iodine

Study the following example to see how we can determine the ethanol content in a brand of spirit using volumetric analysis.



57.1 Determining the concentration of ethanol in a brand of spirit

An experiment is carried out to determine the concentration of ethanol in a brand of spirit. The experiment consists of two stages.

Stage 1	Dilute 25.0 cm^3 of the spirit to 250.0 cm^3. Withdraw 25.0 cm^3 of the diluted spirit. Add 25.0 cm^3 of 0.300 mol dm^{-3} aqueous solution of potassium dichromate and excess dilute sulphuric acid. Allow the mixture to stand at room temperature for about an hour.
Stage 2	Titrate the potassium dichromate remaining in the mixture against 0.200 mol dm^{-3} aqueous solution of iron(II) ions. 15.0 cm^3 of the solution are required to reach the end point.

a) The reaction in Stage 2 can be represented by the following equation:

$$Cr_2O_7^{2-}(aq) + 6Fe^{2+}(aq) + 14H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 6Fe^{3+}(aq) + 7H_2O(l)$$

Calculate the number of moles of potassium dichromate left after the reaction in Stage 1.

b) Calculate the number of moles of potassium dichromate used up in the reaction in Stage 1.

c) The reaction between ethanol in the spirit and potassium dichromate can be represented by the following equation:

$$3CH_3CH_2OH(aq) + 2Cr_2O_7^{2-}(aq) + 16H^+(aq) \longrightarrow 3CH_3COOH(aq) + 4Cr^{3+}(aq) + 11H_2O(l)$$

Calculate the concentration of ethanol in the brand of spirit.

AL 2004 Paper 2 Q4(c)

Solution

a)

$$\underset{\text{? mol}}{Cr_2O_7^{2-}(aq)} + \underset{\substack{0.200 \text{ mol } dm^{-3} \\ 15.0 \text{ } cm^3}}{6Fe^{2+}(aq)} + 14H^+(aq) \longrightarrow 2Cr^{3+}(aq) + 6Fe^{3+}(aq) + 7H_2O(l)$$

Number of moles of Fe^{2+} ions in 15.0 cm^3 solution = 0.200 mol dm^{-3} x $\frac{15.0}{1\ 000}$ dm^3

= 0.00300 mol

According to the equation, 1 mole of $Cr_2O_7^{2-}$ ions reacts with 6 moles of Fe^{2+} ions.

i.e. number of moles of $Cr_2O_7^{2-}$ ions left after the reaction in Stage 1 $= \frac{0.00300}{6}$ mol

$= 5.00 \times 10^{-4}$ mol

b) Number of moles of $Cr_2O_7^{2-}$ ions originally added in Stage 1 $= 0.300 \text{ mol dm}^{-3} \times \frac{25.0}{1\ 000} \text{ dm}^3$

$= 0.00750$ mol

Number of moles of $Cr_2O_7^{2-}$ ions used up in the reaction in Stage 1 $= (0.00750 - 5.00 \times 10^{-4})$ mol

$= 0.00700$ mol

c)

$$3CH_3CH_2OH(aq) + 2Cr_2O_7^{2-}(aq) + 16H^+(aq) \longrightarrow 3CH_3COOH(aq) + 4Cr^{3+}(aq) + 11H_2O(l)$$

? mol dm^{-3} 0.00700 mol

25.0 cm^3

↓

250.0 cm^3

↓

(used) 25.0 cm^3

According to the equation, 3 moles of CH_3CH_2OH react with 2 moles of $Cr_2O_7^{2-}$ ions.

i.e. number of moles of CH_3CH_2OH in 25.0 cm^3 diluted spirit $= \frac{3}{2} \times 0.00700$ mol

$= 0.0105$ mol

Number of moles of CH_3CH_2OH in 250.0 cm^3 diluted spirit

$= 10 \times 0.0105$ mol

$= 0.105$ mol

= number of moles of CH_3CH_2OH in 25.0 cm^3 original spirit

Concentration of CH_3CH_2OH in original spirit $= \frac{0.105 \text{ mol}}{\frac{25.0}{1\ 000} \text{ dm}^3}$

$= 4.20 \text{ mol dm}^{-3}$

$\therefore$ the concentration of ethanol in the spirit is 4.20 mol dm^{-3}.



Fig. 57.2 A chemist using gas chromatography for analysis

Reference Website
Refer to the following website for a video clip on gas chromatography:
http://www.youtube.com/watch?v=08YWhLTjlfo&feature=related
(accessed May 2015)

57.3 Gas chromatography

Gas chromatography is commonly used in food analysis. It uses a gas as the mobile phase, and the stationary phase can either be a solid or a non-volatile liquid. If a solid stationary phase is used, the technique is called **gas-solid chromatography**. If the stationary phase is a liquid, the technique is called **gas-liquid chromatography**. The latter is more commonly used.

Fig. 57.3 shows a schematic diagram of a gas-liquid chromatograph. It has three main parts:

- an **injector**;
- a **chromatographic column** inside an oven (and the stationary phase particles are coated onto the inside of the column);
- a detector which is usually connected to a chart recorder.

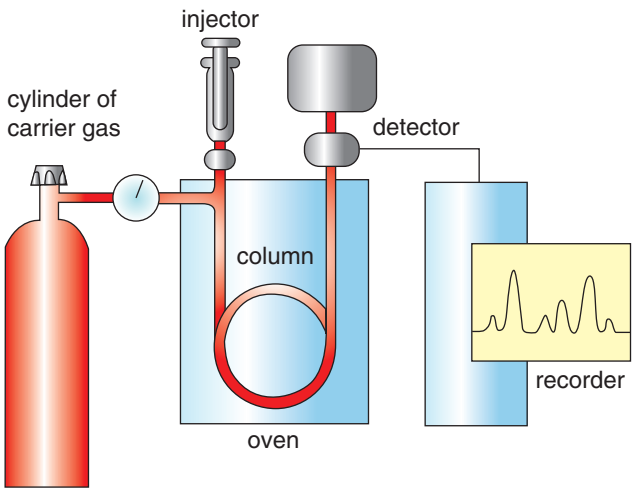


Fig. 57.3 A schematic diagram of a gas-liquid chromatograph

In operation, the sample is injected into the chromatograph via the injector and vaporized using a heater. The sample is swept along the column by a **carrier gas**, usually nitrogen, helium or argon.

Components in the sample that dissolve more readily in the liquid stationary phase move more slowly in the carrier gas stream. Separation of components in the sample results.

After passing through the column, components are detected by a detector. Signals from the detector are plotted out by a recorder as a chromatogram. This shows the recorder response against the time which has elapsed since the sample is injected into the column.

gas-solid chromatography 氣固色層法　gas-liquid chromatography 氣液色層法
injector 注射器　chromatographic column 色層柱　carrier gas 載體氣

Fig. 57.4 shows a typical gas chromatogram of the additives in a soft drink.

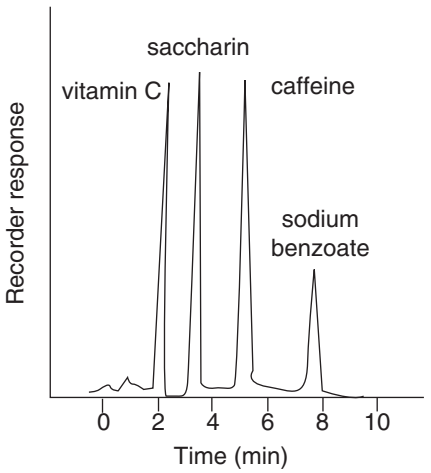


Fig. 57.4 A gas chromatogram of the additives in a soft drink

Qualitative and quantitative analyses

The time a component is held on a column under given conditions is characteristic of the component and is called its **retention time** (symbol: R_t). Identification is made by comparing the retention time of the unknown with retention times of known species.

The area under each peak is proportional to the concentration of that component in the sample. We can obtain the concentration of the component based on the peak area*.

* Peak areas need to be calibrated using a series of standards of the component.

Applications of gas chromatography

This technique is highly sensitive and reliable. *Adulteration* of foods with cheaper alternatives, such as the substitution of methanol for ethanol in wines, has long been a practice of dishonest merchants. Gas chromatography is a powerful tool in detecting such frauds. Quite different chromatograms can result if foodstuffs are tampered with.

N3 Refer to page T89.

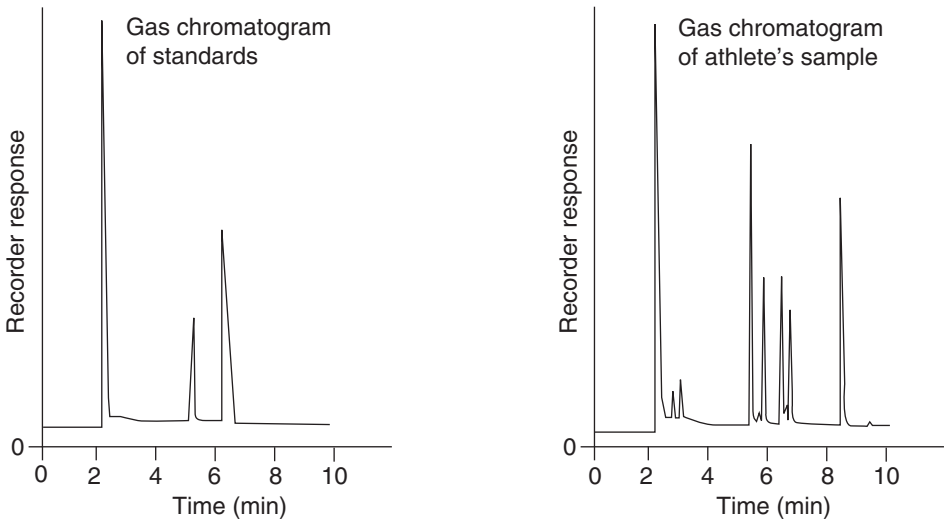
Examples of other uses of gas chromatography include:

- air monitoring in polluted environments, such as gas quality in mines;
- testing of urine samples from athletes for banned substances.

retention time 停留時間　　adulteration 冒充正貨

Checkpoint

A weight lifter is tested for the use of *steroids* using gas chromatography. The resulting chromatogram is compared with the chromatogram of a set of standards.



Based on the chromatograms provided, describe any evidence that the weight lifter has been using steroids. Some peaks in the chromatogram of the athlete's sample coincide with peaks in the standard. It is likely that the weight lifter was using steroids.

57.4 Carbon monoxide in the air

Pollutants in the air

In a modern city, the major air pollutants include carbon monoxide, sulphur dioxide, oxides of nitrogen, etc.

One way of describing the concentrations of pollutants in the air is parts per million (ppm). This is the number of molecules of pollutants per million (10^6) molecules in air. Thus, it is a ratio with no unit.

Extension

Based on the Avogadro's Law, we can also express the concentration of a gas pollutant in the air as follows:

$$\text{Concentration in ppm of a gas pollutant in air} = \frac{\text{volume of gas pollutant}}{\text{volume of air}} \times 10^6 \text{ ppm}$$

steroid 類固醇

Extension

Obviously, the units of volume in the numerator and denominator must agree.

For example, a concentration of 20.0 ppm of carbon monoxide means there are 20.0 cm^3 of the pollutant for every 10^6 cm^3 of air. Nowadays, concentration of a pollutant is more commonly expressed as microgram per cubic metre (μg m^{-3}).



57.2 Expressing the concentration of a gas pollutant in μg m^{-3}

Consider a sample of air containing 0.0200 ppm of sulphur dioxide, measured at room temperature and pressure. Express the concentration of sulphur dioxide in μg m^{-3}.

(Relative atomic masses: O = 16.0, S = 32.1; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

Solution

In 1 m^3 of air, there is $\frac{0.0200}{10^6}$ m^3 of SO_2.

* 1 m^3 = 10^6 cm^3

Molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1}
= 24 000 cm^3 mol^{-1}
= 0.0240 m^3 mol^{-1}*

$$\therefore \text{ number of moles of } SO_2 \text{ in 1 } m^3 \text{ of air} = \frac{2.00 \times 10^{-8}\ m^3}{0.0240\ m^3\ mol^{-1}}$$
$$= 8.33 \times 10^{-7}\ mol$$

Mass of SO_2 in 1 m^3 of air = 8.33 x 10^{-7} mol x 64.1 g mol^{-1}*
= 53.4 x 10^{-6} g
= 53.4 μg

* Molar mass of SO_2 = 64.1 g mol^{-1}.

AL 1999 Paper 2 Q6(d)(i)(I)

∴ the concentration of sulphur dioxide in the sample of air is 53.4 μg m^{-3}.

Extension

Checkpoint

Refer to page T89.

1 The carbon dioxide level in the air of a certain city is 365 ppm, measured at room temperature and pressure. Calculate the concentration of carbon dioxide in the air in μg m^{-3}. 669 000 μg m^{-3}

(Relative atomic masses: C = 12.0, O = 16.0; molar volume of gas at room temperature and pressure = 0.0240 m^3 mol^{-1})

2 The concentration of nitrogen dioxide in a sample of air is 50.0 mg m^{-3}, measured at room temperature and pressure. Calculate the concentration of nitrogen dioxide in the sample in ppm. 26.2 ppm

(Relative atomic masses: N = 14.0, O = 16.0; molar volume of gas at room temperature and pressure = 0.0240 m^3 mol^{-1})

Carbon monoxide in the air of Hong Kong

Carbon monoxide in the air of Hong Kong comes mainly from car emissions, although a small amount may also come from incomplete combustion of fuels from factories and power stations.

Carbon monoxide reacts readily with haemoglobin in the human blood and as a result reduces the oxygen-carrying capacity of the blood when inhaled.

Typical symptoms of carbon monoxide poisoning include shortness of breath, chest pain, headaches and loss of co-ordination.

The concentration of carbon monoxide in the air of Hong Kong remained very low in the past several years (Fig. 57.5).

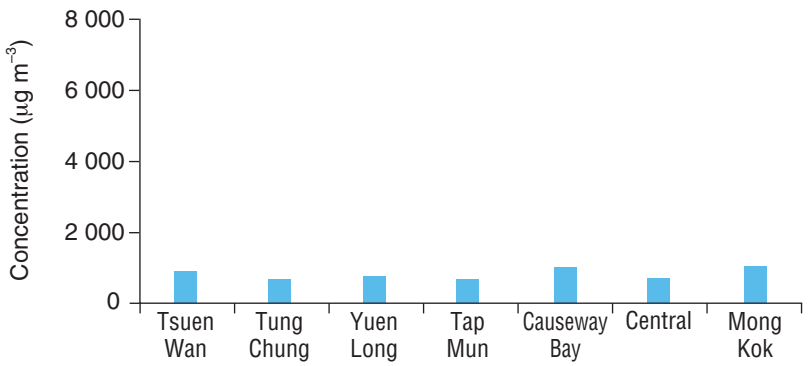


Fig. 57.5 Level of carbon monoxide in various locations in Hong Kong (2013)

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Concentration of carbon monoxide in the air

Carbon monoxide is poisonous. Harmful effects of the pollutant depend on its concentration and the duration of exposure to the pollutant. A healthy adult can tolerate 100 ppm carbon monoxide in the air without suffering ill effects. A one hour exposure to 1 000 ppm causes a headache and a reddish coloration of the skin develops. A one hour exposure to 1 300 ppm turns the skin cherry red and a throbbing headache develops. A one hour exposure to concentrations greater than 2 000 ppm will likely cause death.

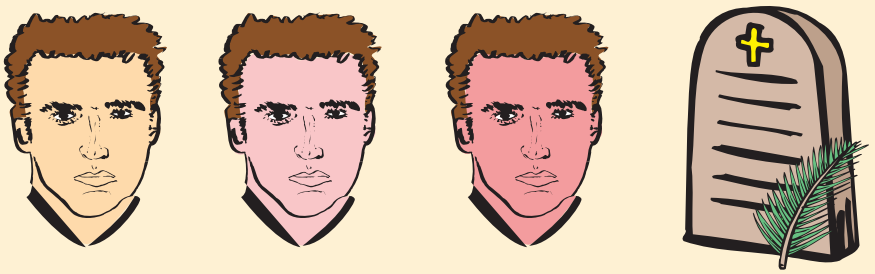


Fig. 57.6 Carbon monoxide toleration of an adult

Determining the concentration of carbon monoxide in the air

The concentration of carbon monoxide in the air can be determined by infrared spectrometry. This technique is based on the fact that carbon monoxide absorbs infrared radiation at certain wavenumbers.

Fig. 57.7 shows the spectrometer designed for determining the concentration of carbon monoxide in the air. It consists of the following parts:

- an infrared source;
- a rotating beam chopper;
- sample and reference cells;
- a detector; and
- an amplifier.

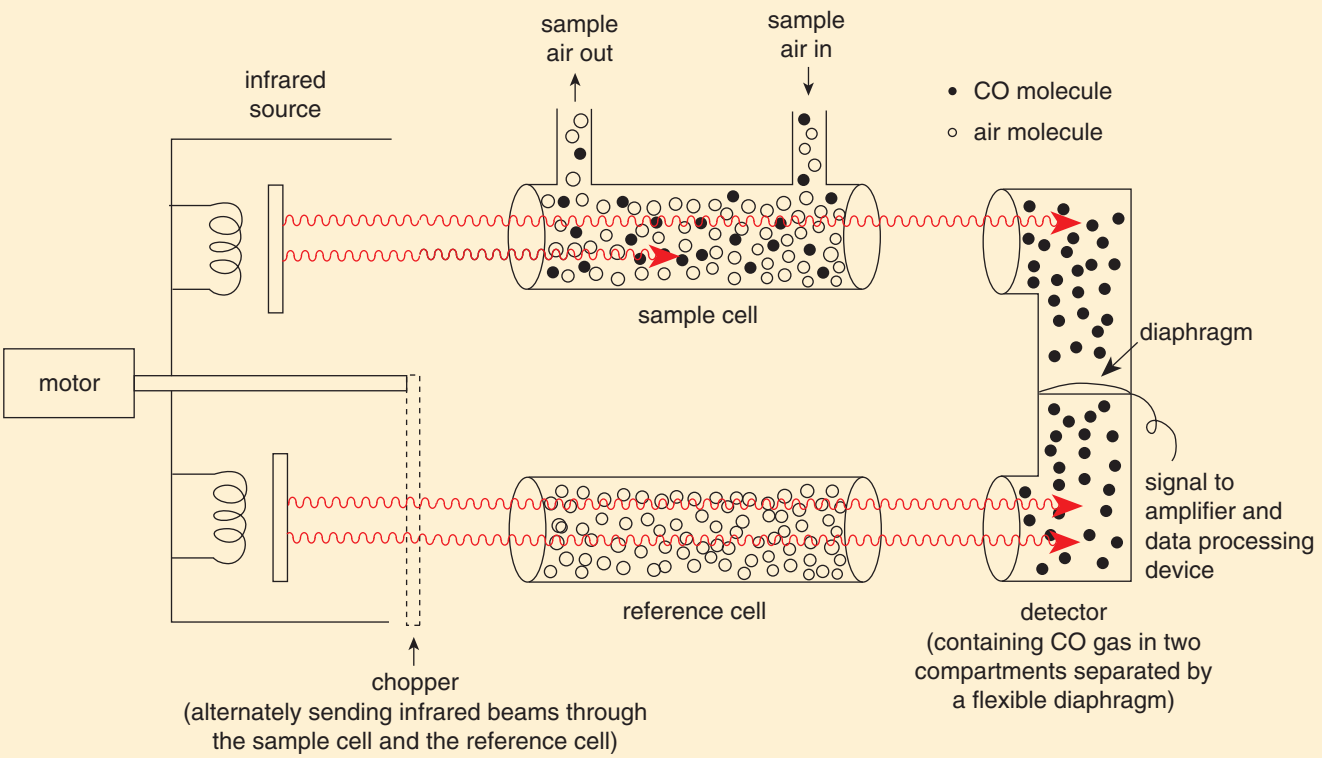


Fig. 57.7 Infrared spectrometer for the determination of carbon monoxide in the air

The rotating chopper allows beams of infrared radiation to pass through the sample cell and reference cell alternately. Both beams then fall on a detector that is filled with carbon monoxide. The detector is separated into two compartments by a flexible diaphragm; the beam passing through the sample cell will fall on one compartment while that passing through the reference cell will fall on the other. The carbon monoxide in both compartments absorbs the infrared radiation falling on it and is heated.

When the beam of infrared radiation passes through the sample cell, some of the infrared radiant energy is absorbed by the carbon monoxide in the sample. Thus, the carbon monoxide in the detector compartment on the sample side receives less infrared radiation and is heated less. This causes the diaphragm to bow towards the sample side. The movement is detected and amplified. The concentration of carbon monoxide in the sample can be determined from a calibration curve prepared using standard carbon monoxide samples.

This system is able to measure carbon monoxide of concentration between 0 ppm to 150 ppm.

Questions

Refer to page T90.

1 *Chronic* exposure to CO may develop *cardiovascular* symptoms in some workers, such as motor vehicle examiners and welders. What deduction can you make if a worker feels especially energetic on holiday?

2 A survey in 2001 indicated that charcoal burning suicide (which leads to CO poisoning) comprised 25% of the suicidal cases in Hong Kong. What measures can the government take to prevent people from committing suicide by burning charcoal?

57.5 Dioxins in the air

DSE 2012 Paper 2 Q3(b)(i)

Dioxins are environmentally persistent organic pollutants comprising **polychlorinated dibenzo-para-dioxins (PCDDs)** and **polychlorinated dibenzofurans (PCDFs)**. They are formed primarily as by-products from the incomplete burning of household and industrial wastes. Exhaust from cars and burning wood also releases dioxins into the air.



PCDDs $x + y = 1 - 8$ PCDFs

Fig. 57.8 Structural formulae of PCDDs and PCDFs

DSE 2012 Paper 2 Q3(b)(ii)

Although the Environmental Protection Agency of the United States has classified dioxins as a probable human carcinogen, there is not sufficient evidence to prove that dioxins cause cancer from exposure to the low levels normally found in the environment*.

According to the Environmental Protection Department, the average concentration of dioxins in the air of Hong Kong is lower than or comparable to levels in large cities in the world.

The four incinerators for treating municipal solid waste in Hong Kong were all closed in the 1990s. However, as the three strategic landfills which currently receive the solid waste will soon be filled up, the Government is considering other methods for bulk waste reduction. Incineration may be a possible method*.

* It has been reported that the dioxins in cigarette smoke (concentration being 18 times higher than the emission standard for modern incinerators) would have more significant impact on human health because cigarette smoke is inhaled directly into the lungs without diffusion and / or dilution.

* Modern incineration plants can be designed and operated to achieve nearly complete destruction of the combustible portion of the waste with very low emission of air pollutants.

chronic 長期的 cardiovascular 心血管的 dioxin 二噁英
polychlorinated dibenzo-para-dioxins (PCDDs) 多氯二聯苯二噁英
polychlorinated dibenzofurans (PCDFs) 多氯二苯呋喃



Fig. 57.9 According to the Government, it is very unlikely that the operation of the incinerator at the Chemical Waste Treatment Centre on Tsing Yi Island will release dioxins at a level that endangers human health

DSE 2012 Paper 2 Q3(b)(iii)

Determining the level of dioxins in the air

N4 The level of dioxins in the air is determined via instrumental analysis but NOT volumetric analysis.

The level of dioxins in the air is very low. This can be measured more accurately via instrumental analysis.

The level of dioxins in the air can be determined using gas chromatography in conjunction with mass spectrometry. This method is known as **gas chromatograph-mass spectrometry (GC-MS)**. It is widely used in environmental pollutant work where the goal is quantitation at very low concentrations in complex mixtures. N4

The basic principle of the GC-MS method is that gas chromatography can separate the species of interest in a complex mixture and a mass spectrometer* can be used as a detector to quantify the species as it elutes from the chromatographic column.

Instead of obtaining a full mass spectrum, the mass spectrometer is set to monitor only m/e values of ions of the species of interest. Then we can obtain the concentration of the species based on peak heights at the m/e values*.

* In Unit 56, we have seen that mass spectrometry can provide detailed structural information on most compounds. The tool can provide quantitative information as well.

* Peak heights are first calibrated with known amounts of the species of interest injected into the GC-MS in a different experiment.

gas chromatograph-mass spectrometry (GC-MS) 氣相色譜儀-質譜法

57.6 Formaldehyde — a common indoor air pollutant

Nowadays most people are aware of the damage that outdoor air pollution can do to their health, but many do not know that indoor air pollution can also have significant health effects.

Formaldehyde (methanal, HCHO) is a common indoor air pollutant. Levels of formaldehyde are usually higher indoors than outdoors because there are many indoor sources of formaldehyde.

The predominant emission sources of formaldehyde are pressed-wood products made with **urea-formaldehyde resins**. These resins are used as adhesives in materials such as *particle board*, *medium-density fibreboard* and *plywood*.



particle board　　medium-density fibreboard　　plywood paneling

Fig. 57.10 Several types of wood materials use urea-formaldehyde resins as adhesives

Formaldehyde is also emitted from cigarette smoking and combustion sources such as gas appliances and kerosene heaters.

Formaldehyde is a colourless gas with a pungent odour at high concentration. It is highly irritating to the eyes and respiratory tract. Breathing air containing low levels of formaldehyde can cause burning and watery eyes. High levels of formaldehyde can cause burning of the nose and throat, coughing and difficulty in breathing.

Determining the level of formaldehyde in indoor air

To determine the level of formaldehyde in indoor air, expose a sampler to the air for a certain time period. The formaldehyde vapour migrates into the sampler and is collected on a tape which has been coated with 2,4-dinitrophenylhydrazine and phosphoric acid. A stable hydrazone forms. Then analyze a solution of the hydrazone with a **high performance liquid chromatograph (HPLC)**.

formaldehyde 甲醛　　urea-formaldehyde resin 脲甲醛樹脂　　particle board 粒子板
medium-density fibreboard 中等密度纖維板　　plywood 膠合板
high performance liquid chromatograph (HPLC) 高效能液相色譜儀

In a high performance liquid chromatograph, the components in a mixture are separated by passing a mobile liquid phase through a stationary phase under high pressure. We can obtain the concentration of formaldehyde based on the peak area of the corresponding signal.



Fig. 57.11 A high performance liquid chromatograph

High performance liquid chromatography is a powerful analytical technique for the identification and quantification of species in complex mixtures.

57.7 Forensic chemistry

Few processes are more important to society than solving crimes, both to protect the public from criminals and to protect the innocent from unjust punishment.



Fig. 57.12 The police working in a crime scene

Crimes investigated by a *forensic chemist* may be crimes by or against individuals, for example, theft, drink driving, *counterfeiting*, *arson* and murder, etc. Forensic chemists are also involved in the investigation of crimes against the society, for example, food adulteration, environmental pollution and use of unsafe chemicals.

For example, arsonists often use accelerants such as gasoline or kerosene to speed the combustion and spread flames. Accelerants used can be identified readily using gas chromatography. Fig. 57.13 shows gas chromatograms of a certain brand of gasoline and vapours from soil recovered from an arson site. Notice their similarity.

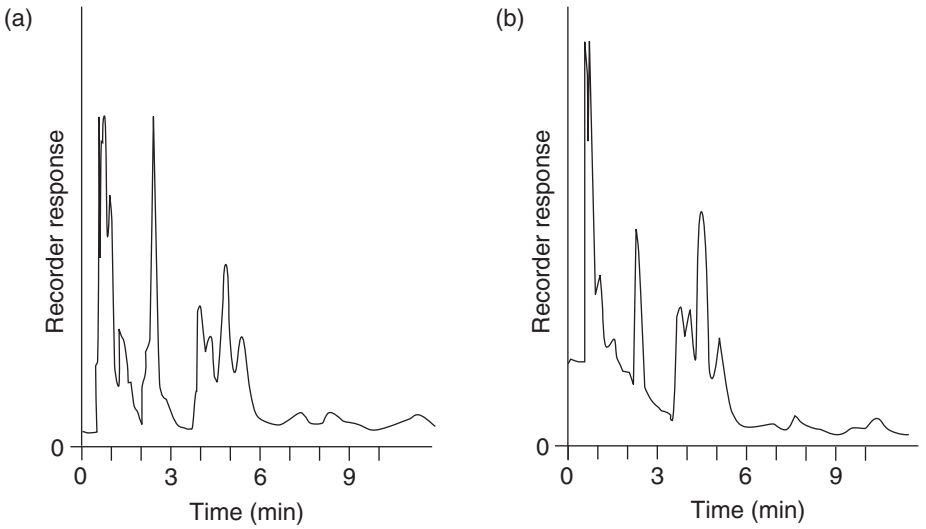


Fig. 57.13 (a) Gas chromatogram of a certain brand of gasoline
(b) Gas chromatogram of vapours from the soil recovered from an arson site

Because of the uniqueness of the infrared spectra of most species, it is often possible to identify a sample by comparing its infrared spectrum with standard spectra. Fig. 57.14 shows the spectra of some polymers. A sample of a polymer at a crime scene can be identified by comparing its infrared spectrum with standard spectra such as those in Fig. 57.14.

forensic chemist 法醫化學專家　　counterfeiting 偽造　　arson 縱火

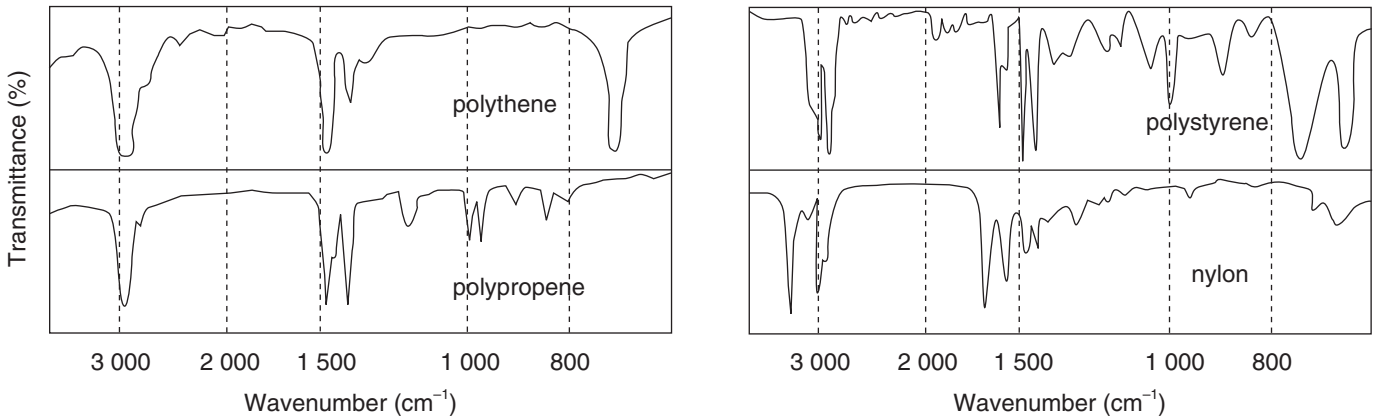


Fig. 57.14 Infrared spectra of polythene, polypropene, polystyrene, and nylon

57.8 Drink driving



Fig. 57.15 Warning symbol 'Don't drive after drinking'

After a person has taken an alcoholic drink, the alcohol passes quickly through the walls of the stomach and intestines into the blood vessels. It is then carried to all parts of the body, including the heart, the brain and muscles.

The ethanol in alcoholic drinks affects the nervous system and even in small amounts can reduce judgement and driving skills.

Blood alcohol content (BAC) is closely related to the extent of effects of alcoholic drinks.

BAC = milligrams of ethanol per 100 cm^3 of blood

The content of ethanol in the blood rises for some time after taking a drink. Then it slowly decreases as the ethanol is excreted or *metabolized*.

The driving limit of blood alcohol content in Hong Kong is 50 mg of ethanol per 100 cm^3 of blood*.

Gas-liquid chromatography is used to identify exactly how much ethanol is present in the blood or urine of a suspect driver. As it is very accurate, the results can be used as evidence in court cases.

* Blood alcohol content may also be expressed as a percentage. Some communities have set a legal limit of 0.080%, meaning that 100 cm^3 of blood would contain 0.080 g of ethanol.

blood alcohol content (BAC) 血液中酒精濃度　　metabolize 使新陳代謝

Breath alcohol testing

Alcohol taken by a person is not chemically changed in the bloodstream. As the blood goes through the lungs, some of the alcohol moves across the membranes of the lung's air sacs into the air. The alcohol content in this air is related to the blood alcohol content. As the alcohol in this air is exhaled, it can be detected by a breath alcohol testing device*.

A police officer can test the driver's breath on the spot and instantly know if there is a reason to arrest the driver.

* The ratio of breath alcohol to blood alcohol is 2100:1. This means that 2100 cm^3 of exhaled air will contain the same amount of alcohol as 1 cm^3 of blood.

N5 Refer to page T91.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

The working principle of a breathalyzer

The technology of breath alcohol testing has changed fundamentally over the years. This was partly driven by general technology advancements and partly due to defense challenges. N5

The first significant breakthrough in evidential breath testing technology began with the Breathalyzer® 900 in 1954. It paved the way for the acceptance of breath alcohol testing by the international forensic community and the courts.



Fig. 57.16 A driver taking a breath alcohol test

When orange dichromate ions ($Cr_2O_7^{2-}$) are brought into contact with ethanol, a redox reaction will occur and green chromium(III) ions (Cr^{3+}) will form. This colour change is the basis of a breathalyzer.

The figure below shows the simplified diagram of a breathalyzer. It has three main parts:

- a system to sample the breath of the suspect;
- two small glass bottles containing the reaction mixture (the test bottle and the reference bottle);
- a system of photocells connected to a meter to measure the colour change associated with the chemical reaction.

CE 2002 Paper 1 Q3(c)
CE 2004 Paper 1 Q8(c)(i)

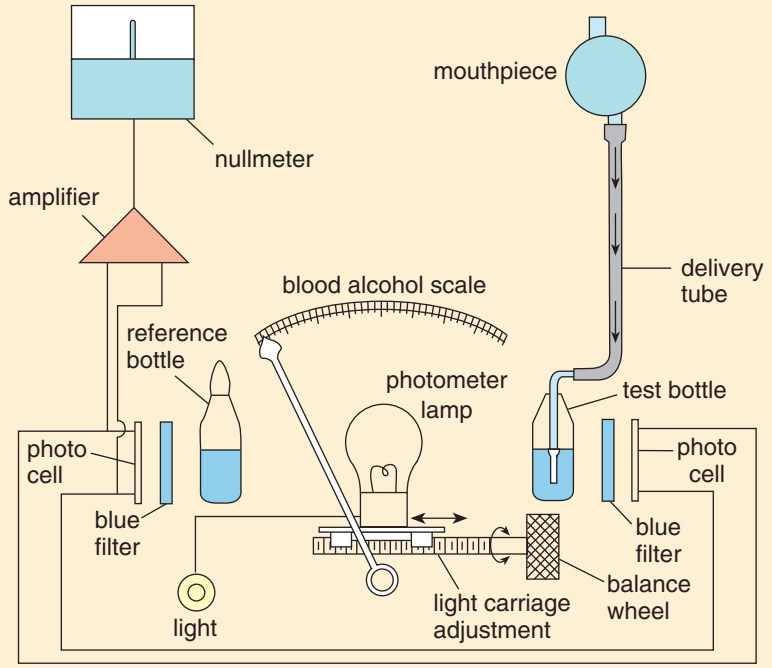


Fig. 57.17 Components of a breathalyzer

To measure the breath alcohol content, a suspect breathes into the device. The breath sample is bubbled through a mixture of potassium dichromate, sulphuric acid and water in the test bottle. The following chemical reaction takes place:

$$2Cr_2O_7^{2-}(aq) + 3CH_3CH_2OH(aq) + 16H^+(aq) \longrightarrow 4Cr^{3+}(aq) + 3CH_3COOH(aq) + 11H_2O(l)$$

To determine the alcohol content of the exhaled air, the device compares identical light beams which are shone through the reacted mixture and the unreacted mixture in the reference bottle. An electric current develops if their intensities are different. This causes the needle of the nullmeter to move from its resting place. The operator then rotates the balance wheel to balance the transmitted light so as to bring the needle of the nullmeter back to the resting place. This also causes a pointer on a scale to move and indicate the blood alcohol content of the person being tested.

Questions

Refer to page T91.

1 When a policeman is using a breathalyzer to test the breath of a driver, the driver claims that he has just rinsed his mouth with ethanol-containing mouthwash. How can the policeman check whether the driver's claim is valid?

2 Should drinking be restricted?

3 Should the government ban television alcohol advertising?

CE 2004 Paper 1 Q8(c)(ii)

In the following activity, you are going to design and make a portable alcohol breathalyzer based on the redox reaction involving potassium dichromate.

Assessment Form

An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.



Activity 57.1

Designing and making a portable alcohol breathalyzer.

57.9 Fingerprint analysis

Fingerprinting is the most cost effective method of suspect identification. The fingers of a person contain small ridges. The ridges form a pattern that is unique to each individual.

There are three types of fingerprints that exist at crime scenes:

- visible prints are made from a finger stained with coloured materials such as ink, blood and grease;
- plastic prints are formed by pressing onto a soft surface such as clay, soap and wax;
- latent prints are invisible prints left on an object by the body's natural greases and oils.

Because a latent print cannot be seen by naked eyes, fingerprint powders, chemicals and even lasers are used to make it visible.

During an investigation, the police compare fingerprints obtained from the scene of a crime against database of fingerprints of known criminals. This may provide the police with a named individual.

Developing fingerprints — fuming with iodine

Latent fingerprints can be developed with iodine fumes. This process relies on the absorption of iodine vapour by the lipids present in the fingerprint deposit to give a yellowish brown coloration.

Fingerprints developed with iodine fade quickly and usually show poor contrast. It is therefore advantageous to fix the prints chemically by the application of a reagent, such as *benzoflavone* which turns the print dark blue.

benzoflavone 苯並黃酮



Fig. 57.18 A fingerprint has been developed using iodine fuming; half of the print has been fixed using benzoflavone

Fig. 57.18 shows a print that has been developed using iodine. Half of the print has been fixed using benzoflavone.

The technique may be applied to a wide range of surfaces such as paper, wood, plastic and glass.

57.10 The role of analytical chemistry in clinical diagnoses

Analytical chemistry is the basis for clinical laboratory tests that measure concentrations of chemicals, principally in blood and urine. Such information helps doctors in the diagnosis, monitoring and prevention of diseases.

Blood is a complex fluid containing many inorganic ions, proteins, carbohydrates, lipids, hormones and enzymes, along with two dissolved gases, oxygen and carbon dioxide. For a healthy person, the concentration of each substance in blood is maintained within limits that reflect normal body metabolism. Disease is often associated with one or more disturbances in this balance of blood chemistry. It is this general principle that underlies the importance of chemical testing of blood in the diagnostic process.



Fig. 57.19 Diabetics may be required to self-check their blood glucose levels every day

For example, the blood glucose testing is carried out to measure the amount of glucose in the blood. This helps in the diagnosis, monitoring and prevention of *diabetes*.

With known diabetics, the doctor will order a glucose test at regular intervals to monitor the patients' status. Diabetics may be required to self-check their glucose every day to monitor glucose levels and to determine if the diabetes treatment plans are working.

Blood glucose testing can also be used to screen healthy, *asymptomatic* individuals for diabetes and pre-diabetes. Pre-diabetes is a condition in which blood glucose levels are higher than normal but not yet diabetic. People with pre-diabetes are at increased risk for developing diabetes and for heart disease and stroke. With modest weight loss and moderate physical activity, people can delay or prevent diabetes.

diabetes 糖尿病　　asymptomatic 無症狀

Non-invasive blood glucose meter

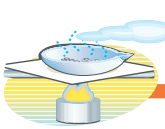
Current blood glucose level monitoring devices require a person to *prick* a finger to draw blood for analysis. Recently, researchers of the Hong Kong Polytechnic University have developed a 'non-invasive blood glucose meter' using near-infrared technology to measure blood glucose level and have been awarded 'Gold Medal with Jury's Congratulations' in an international invention exposition held in Geneva.

These researchers have identified near-infrared radiation to be suitable for monitoring blood glucose level. When a diabetic puts his/her finger on the meter's detector, blood glucose can absorb near-infrared radiation shone on the diabetic's finger. The blood glucose level can be obtained by sopisticated mathematical calculations based on the parts of radiation absorbed and reflected. The patient experiences no pain as no prick is needed.



(Source: JKP 05/2007)

prick 戳破



Summary

1 Analytical chemistry serves the needs of many fields including food and drug analyses, environmental quality control, forensic chemistry and clinical laboratory tests.

2 We can employ volumetric analysis or gas chromatography to determine the ethanol content of a spirit.

3 a) Carbon monoxide level in the air may be determined by infrared spectrometry.

b) The level of dioxins in the air may be determined by gas chromatograph-mass spectrometry (GC-MS).

c) Formaldehyde is a common indoor pollutant. The level of formaldehyde in indoor air may be determined by high performance liquid chromatography (HPLC).

4 a) The working of an alcohol breathalyzer is based on the redox reaction involving potassium dichromate.

b) Latent fingerprints can be developed with iodine fumes.

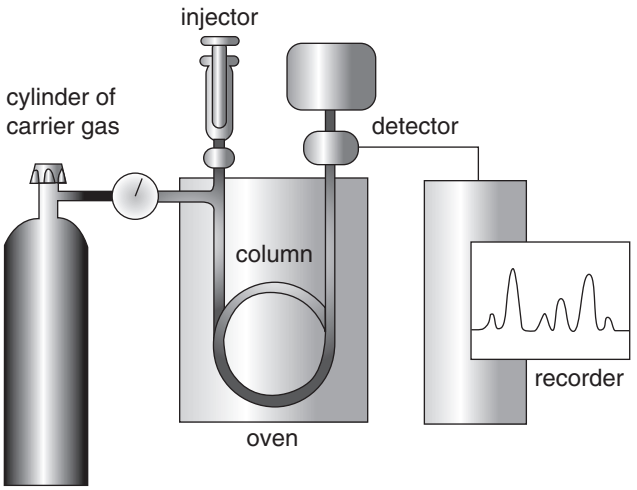
5 Analytical chemistry is also the basis for clinical laboratory tests that measure concentrations of chemicals, such as glucose, principally in blood and urine.

Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

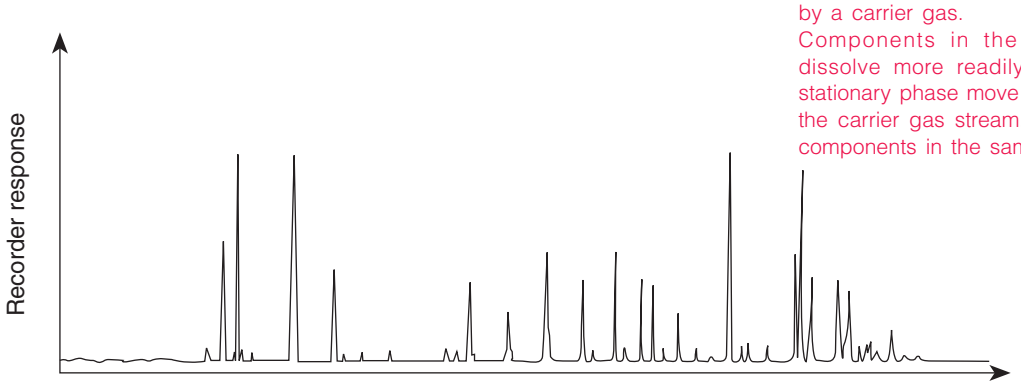
1 The diagram below shows a gas chromatograph.



a) Identify the mobile and stationary phases in the gas chromatograph shown and explain how gas chromatography is able to separate the components of a mixture.

a) The mobile phase is the carrier gas. The stationary phase particles are coated onto the inside of the column. The sample is swept along the column by a carrier gas.
Components in the sample that dissolve more readily in the liquid stationary phase move more slowly in the carrier gas stream. Separation of components in the sample results.

b) The gas chromatograph of a premium grade petrol is shown below.



What is plotted on the horizontal axis?

Time

c) Suggest ONE use of gas chromatography.

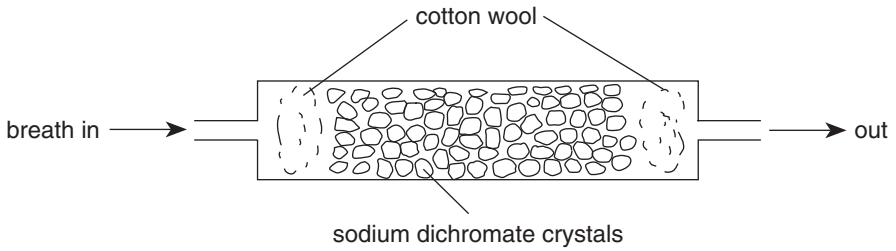
Any one of the following:
- Checking whether foodstuffs are tempered with cheaper alternatives.
- Air monitoring in polluted environments, such as gas quality in mines.
- Testing of urine samples from athletes for banned substances.
- Determining the concentration of a certain component in a sample.

2 Dioxins are persistent organic pollutants.
Refer to page T92.

a) Suggest a source of dioxins.

b) The level of dioxins in the air can be determined using gas chromatography in conjunction with mass spectrometry (GC-MS). Briefly describe how this works.

3 The diagram below shows a breathalyzer made by a student.
Refer to page T92.



What will be observed when the breath of a drunken person is passed through the breathalyzer? Explain the observation with the help of an equation.

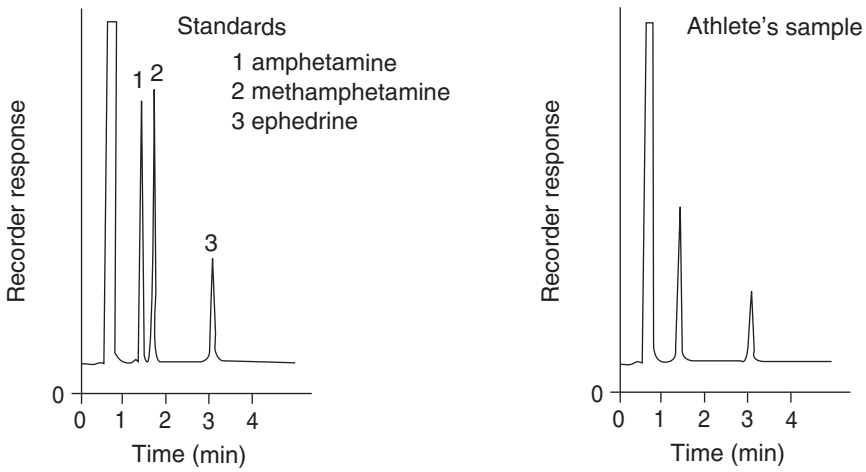
4 Suggest a method of making latent fingerprints visible. Briefly describe the chemistry behind the method. Develop latent fingerprints with iodine fumes.
The lipids present in the fingerprint deposit give a yellow brown coloration upon the absorption of iodine vapour.

5 Suggest ONE application of analytical chemistry in clinical diagnoses. Any one of the following:
- Blood glucose test
- Blood cholesterol test
- Phosphorus content in urine

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 An athlete is tested for amphetamines which are sometimes used to enhance performance. From the gas chromatograms below, what evidence is there that the athlete tested has been using amphetamines?
One peak in the chromatogram of the athlete's sample coincides with that of amphetamine.



7 Chromatography is an important analytical technique used to separate mixtures into their individual components.

Refer to page T93.

a) Define the following terms used in chromatography.

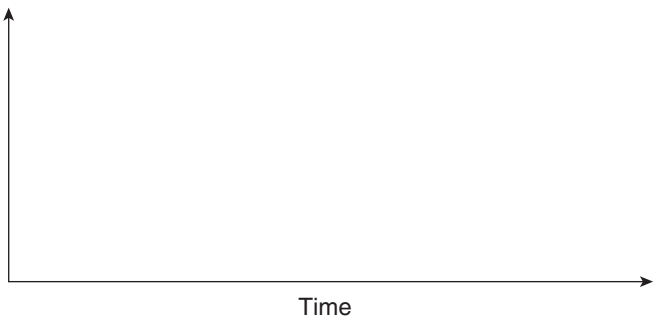
i) R_f value

ii) Retention time

b) Volatile organic compounds that enter soil via waste disposal sites affect the quality of the soil. A preliminary step in analyzing soil quality involves separation of these volatile components using gas-liquid chromatography.

i) What are the roles of the gas and liquid in gas-liquid chromatography?

ii) Draw a diagram of a gas-liquid chromatogram for a mixture containing two components.



iii) Explain how the gas-liquid chromatogram could be used to determine the percentage composition of each component in the mixture.

(OCR Advanced GCE, Methods of Analysis and Detection, Jan. 2007)

8 a) Under acidic conditions, ethanol reacts with dichromate ions quantitatively to give ethanoic acid and chromium(III) ions.

Write an equation for the reaction of ethanol with dichromate ions under acidic conditions.

b) In an experiment to determine the concentration of ethanol in a brand of spirit, 10.0 cm^3 of a sample of the spirit was diluted to 250.0 cm^3. 25.0 cm^3 portions of the diluted spirit were withdrawn. To each portion, 25.0 cm^3 of 0.156 M $K_2Cr_2O_7(aq)$ and excess dilute H_2SO_4 were added. The mixtures obtained were allowed to stand at room temperature for about an hour. The excess $K_2Cr_2O_7$ in each mixture was then titrated against 0.118 M $(NH_4)_2Fe(SO_4)_2(aq)$ with an appropriate indicator. The mean titre was 12.23 cm^3.

Calculate the concentration of ethanol, in mol dm^{-3}, in this brand of spirit.

(HKALE 2004)

9 The sulphur dioxide content of white wine can be determined by titration. A sample of white wine is titrated against a standard iodine solution from a burette, using starch solution as indicator.

a) Write a chemical equation for the reaction involved in this titration.

b) State the colour change at the end point.

(HKASLE 2006)

Extension

10 A sample of air contains 29 900 μg m^{-3} of nitrogen dioxide, measured at room temperature and pressure.

15.6 ppm; refer to page T93.

Express the concentration of nitrogen dioxide in ppm by volume.

(Relative atomic masses: N = 14.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

11 The World Health Organization (WHO) adopts the following guidelines for maximum carbon monoxide concentrations and periods of exposure:

Refer to page T93.

- 90 ppm by volume for 15 minutes;
- 50 ppm by volume for 30 minutes;
- 25 ppm by volume for 1 hour.

a) What does the unit 'ppm' stand for?

b) A sample of car exhaust contains 12 ppm of carbon monoxide (measured at room temperature and pressure). Calculate the concentration of carbon monoxide, in mg m^{-3}, in the sample. 14 mg m^{-3}

c) Explain why car exhaust contains carbon monoxide.

d) State ONE health hazard associated with carbon monoxide.

12 An experiment is carried out to determine the SO_2 level in the air. The experiment consists of two steps.

Refer to page T94.

Step 1 Bubble 240 dm^3 of an air sample, measured at room temperature and pressure, into excess $H_2O_2(aq)$. The $H_2O_2(aq)$ converts the SO_2 into $SO_4^{2-}(aq)$.

Step 2 Treat the resulting solution with excess $BaCl_2(aq)$ to give $BaSO_4(s)$.

0.0700 g of $BaSO_4(s)$ is obtained in the experiment.

Calculate the concentration of SO_2 in the air sample, expressing your answer in

a) μg m^{-3}; and 8.00×10^4 μg m^{-3}

b) ppm by volume. 30.0 ppm

(Relative atomic masses: O = 16.0, S = 32.1, Ba = 137.3; molar volume of gas at room temperature and pressure = 24.0 dm^3 mol^{-1})

13 Formaldehyde (methanal) is a common indoor air pollutant.

Refer to page T94.

a) Suggest an indoor source of the pollutant.

Extension

b) The indoor air quality (IAQ) standard for formaldehyde has been set at 0.10 ppm by volume.

Calculate the maximum allowable quantity of formaldehyde, in grams, in a room of volume 360 m^3 (measured at room temperature and pressure). 0.045 g

c) State ONE health hazard associated with formaldehyde.

d) Name an instrument that can be used to determine the level of formaldehyde in indoor air.

(Relative atomic masses: H = 1.0, C = 12.0, O = 16.0; molar volume of gas at room temperature and pressure = 24.0 $dm^3\ mol^{-1}$)

14 Ozone is important not only for its protective effects from harm caused by ultraviolet radiation, but also for water and sewage treatment. It is usually produced as a mixture with oxygen called ozonized oxygen.

Refer to page T95.

The amount of ozone in ozonized oxygen can be determined by using the ozone to oxidize iodide ions and titrating the liberated iodine against standard aqueous solution of sodium thiosulphate.

$$O_3(g) + 2I^-(aq) + H_2O(l) \longrightarrow O_2(g) + I_2(aq) + 2OH^-(aq)$$

10.0 dm^3 of ozonized oxygen were shaken repeatedly with an aqueous solution of potassium iodide. The liberated iodine required 40.0 cm^3 of 1.00 mol dm^{-3} sodium thiosulphate solution for reduction.

a) Write an ionic equation for the reaction between the liberated iodine and the aqueous solution of sodium thiosulphate.

b) Briefly describe how the end point of the titration can be determined.

c) Calculate the percentage by volume of ozone in the ozonized oxygen. 4.80%

(Molar volume of gas at temperature and pressure of the experiment = 24.0 $dm^3\ mol^{-1}$)

15 Sulphur dioxide is often added to wine as a preservative, but it is important to get the final concentration correct. If too much is added the wine tastes of sulphur dioxide, too little and the wine goes bad. The concentration of sulphur dioxide in wine can be calculated by titrating the wine with an aqueous solution of iodine.

Refer to page T95.

a) The equation for the reaction between sulphur dioxide and iodine is shown below.

$$SO_2(aq) + I_2(aq) + 2H_2O(l) \longrightarrow SO_4^{2-}(aq) + 2I^-(aq) + 4H^+(aq) \qquad \textbf{equation 1}$$

i) Give the oxidation states of the iodine and sulphur before and after the reaction, by completing the following table.

Oxidation state of sulphur in:	Oxidation state of iodine in:
SO_2 = +4	I_2 = 0
SO_4^{2-} = +6	I^- = –1

ii) State, with a reason, whether sulphur is oxidized or reduced in the conversion of SO_2 into SO_4^{2-}.

b) A sample of white wine was analyzed for its sulphur dioxide content by titrating it with an aqueous solution of iodine.

i) State the colour change of the iodine as it reacts with the sulphur dioxide solution.

ii) 16.20 cm^3 of 0.0100 mol dm^{-3} aqueous I_2 solution was needed to react with 50.00 cm^3 of the wine. Calculate the number of moles of iodine, I_2, used in the titration. 1.62×10^{-4} mol

iii) Use **equation 1** to write down the number of moles of sulphur dioxide in the 50.00 cm^3 of wine. 1.62×10^{-4} mol

iv) What is the concentration of sulphur dioxide in the wine in mol dm^{-3}? 3.24×10^{-3} mol dm^{-3}

v) The concentration of sulphur dioxide in wine is usually quoted in g dm^{-3}. What is the concentration of sulphur dioxide in this wine in g dm^{-3}? 0.208 g dm^{-3}

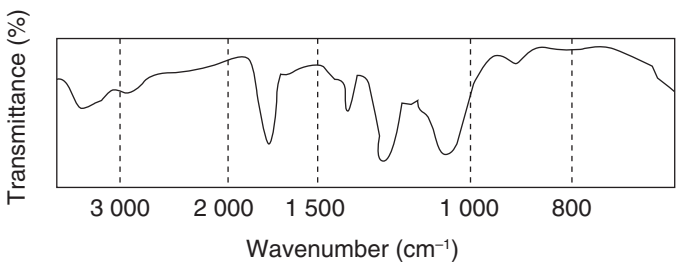
(Relative atomic masses: O =16.0, S = 32.1)

vi) The generally accepted maximum concentration of sulphur dioxide in wine is 0.25 g dm^{-3}. A concentration of less than 0.01 g dm^{-3} is insufficient to preserve the wine. Comment on the effectiveness of the sulphur dioxide in the wine analyzed in (v).

(OCR Advanced Subsidiary GCE (Salters), Chemistry of Natural Resources, Jun. 2007)

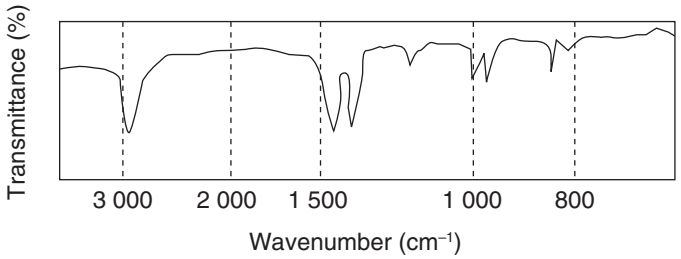
16 In order to determine the chemical nature of (a) an overhead projector slide, and (b) a plastic grocery bag, an analyst obtained the infrared spectrum of each. Use the set of standard spectra shown below to identify the chemical nature of (a) and (b).

a) Infrared spectrum of an overhead projector slide
Cellulose acetate



b) Infrared spectrum of a plastic grocery bag

Polypropene



Standard spectra

